JugglAR: Exploring Mixed Reality for Enhanced Task Learning and Muscle Memory Formation

HARSHVARDHAN CHANDIRASEKAR*, University of Minnesota, USA
AMANDA ELVARSDÓTTIR*, University of Minnesota, USA
ANURAG PUROHIT, University of Minnesota, USA
DYLAN BLUNDELL, University of Minnesota, USA
ZHU-TIAN CHEN, University of Minnesota, USA

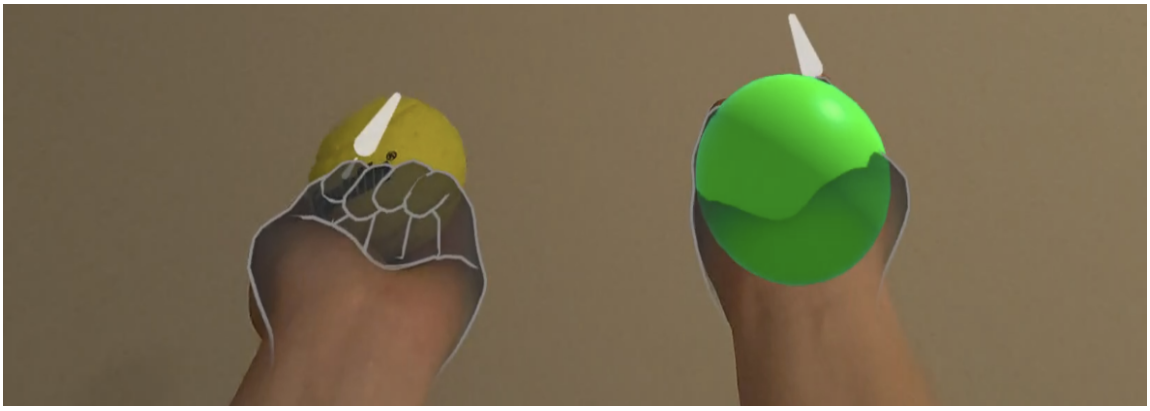


Fig. 1. A Tennis Ball vs. JugglAR Ball, 2025.

Augmented Reality (AR) applications increasingly leverage muscle memory to enhance user interaction with virtual objects. However, it remains unclear how this learned interaction transfers to the physical world. This study investigates how AR-based muscle memory translates to physical tasks, using juggling as a case study. We developed JugglAR, an AR application replicating physical juggling with hand-tracking and adjustable settings for ball speed and gravity, to create a customizable learning environment. Two groups, one trained in physical juggling and the other using JugglAR, were compared to evaluate skill acquisition, efficiency, and retention when transitioning to physical juggling. Our findings demonstrate AR's potential as an effective tool for teaching complex motor skills, offering an accessible and adaptable platform for skill training. Insights gained from this research highlight AR's role in bridging virtual and physical learning experiences, with implications for education and training.

*Both authors contributed equally to this research.

Authors' Contact Information: Harshvardhan Chandirasekar, University of Minnesota, Minneapolis, Minnesota, USA; Amanda Elvarsdóttir, University of Minnesota, Minneapolis, Minnesota, USA; Anurag Purohit, University of Minnesota, Minneapolis, Minnesota, USA; Dylan Blundell, University of Minnesota, Minneapolis, Minnesota, USA; Zhu-Tian Chen, University of Minnesota, Minneapolis, Minnesota, USA.

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CCS Concepts: • **Human-centered computing** → **User studies**; **Mixed / augmented reality**; **Gestural input**; • **Computing methodologies** → *Mixed / augmented reality*.

Additional Key Words and Phrases: Augmented Reality (AR), Motor Skills, Muscle Memory, Juggling

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1 Introduction

Muscle memory plays a crucial role in developing coordination, accuracy, and efficiency for complex motor skills. This memory is established through consistency and repetition [20], which can be challenging to achieve due to factors like speed, trajectory, and gravity. To address these challenges, our research introduces an augmented reality (AR) application that allows users to interact with virtual juggling balls, mimicking the physics of real juggling while offering additional controls. Virtual environments have been shown to play a significant role in training motor skills across various domains, including sports [1].

Juggling was chosen as the focus of our study because it is a representative motor skill that demands coordination, hand-eye synchronization, and trajectory prediction through precise, repetitive actions. It is also a widely accessible skill that can be practiced with simple materials such as balls [15]. Juggling is lightweight and requires minimal physical strain, making it a suitable task for AR simulations. We hypothesize that the skills acquired in AR can transfer seamlessly to real-world practice, preserving muscle memory across both environments.

However, learning juggling in the physical world presents several challenges. Beginners often experience disruptions from repeatedly picking up dropped balls, which can lead to frustration and reduced practice time. The fast pace of juggling further complicates the ability to track multiple objects simultaneously. As noted in [14], these challenges can hinder sustained practice, especially for individuals who might benefit from learning at a slower pace. Adjusting parameters like ball speed and gravity, as enabled in our AR application, could help learners focus better and improve their outcomes.

Building upon previous research, we aim to address these challenges through a hand-tracked AR application for virtual motor skill learning. Juggling 4.0, for example, augmented physical juggling with visual feedback, using a Microsoft Kinect and HoloLens to help users throw a single ball to a consistent height. However, it only supported limited physical movements and did not train complete juggling skills. Similarly, Juggling in VR demonstrated how virtual reality could enhance motivation and provide effective training by incorporating reduced gravity, trajectory lines, and haptic feedback via controllers. While effective, it relied on VR controllers and lacked realistic hand-tracking.

Our research builds on these foundations by introducing JugglAR, a hand-tracked AR application designed to train users in realistic juggling movements. By replicating catching and releasing motions, we aim to develop proper muscle memory that transfers effectively to physical juggling. The key contributions of this work are: (1) an AR application for realistic juggling training using hand-tracking technology, and (2) an evaluation of its effectiveness through a user study, analyzing its limitations and potential as a learning tool for novice jugglers.

2 Related Work

2.1 Virtual Reality Environments and Tasks

Virtual reality environments have shown great promise in improving motor skills and rehabilitation in various fields, including golf [10], baseball [6, 13], industrial maintenance [8], table tennis [3, 18], and even juggling [5]. These studies have demonstrated the potential of different virtual reality techniques, such as visual, audio, and haptic, to improve skill learning significantly. Virtual reality can create consistent training scenarios that imitate the physical demands of real tasks and develop the necessary motor skills. [16]. There is also an emphasis on the user's ability to modify a virtual environment to optimize their interactive experience or better immerse themselves.

However, the use of virtual environments for training has its challenges. These include the development of practical, complex training tasks and environments with accurate physics models and ensuring real-time responses [17]. For example, a virtual juggling trainer may struggle to determine the subtle differences between correct and incorrect catches and throws. Another challenge is to select the best output modality to convey specific information and engage the user while reducing the cognitive load [5]. Despite these challenges, their management can transform virtual environments into a powerful tool for learning new skills.

Hand-tracking has become a promising method of interaction to increase the immersion of virtual environments for skill training. It increases the sense of presence and immersion in the environment by allowing natural direct manipulation with virtual objects. Hameed et al. [9] compared hand-tracking to traditional handheld controllers in 2021, finding that while hand-tracking can make training immersive, it can also increase the mental load of training. However, growing familiarity and recent advances in hand-tracking research may mitigate this and other limitations related to the accuracy and robustness of the interaction modality.

2.2 Juggling-Specific Research

Research emphasizes the motor skills and learning process needed to become proficient in juggling. Behind this skill is an understanding of physics and math, such as dwell time (the duration a ball remains in the hand), flight time (the time a ball spends in the air), and hand cycles (the coordination of throwing and catching patterns), as explored by Beek et al. [2]. A dynamic systems perspective of the learning process identifies stages characterized by the layout, which could increase stability and coordination for the user [11].

The role of practice in improving juggling performance is well-documented. Bebko et al. [12] found that consistent practice leads to significant improvements in both performance and automaticity, as learners refine motor patterns through repetition. Furthermore, eye-tracking studies reveal striking differences between novice and expert jugglers. While beginners tend to track individual balls with their eyes, experts adopt a more efficient visual strategy, focusing on a central point and relying on peripheral vision to monitor ball trajectories [19]. This centralization of visual attention plays a crucial role in improving stability and accuracy, underscoring the importance of visual strategies in developing juggling proficiency.

These insights have important implications for the design of virtual juggling applications. To facilitate efficient learning, it is essential to account for visual attention and optimize the virtual environment to encourage proper movement patterns and perceptual strategies. By simulating realistic ball trajectories and providing visual cues, such applications can support learners in building the foundational motor and cognitive skills necessary for successful juggling.

3 System Design

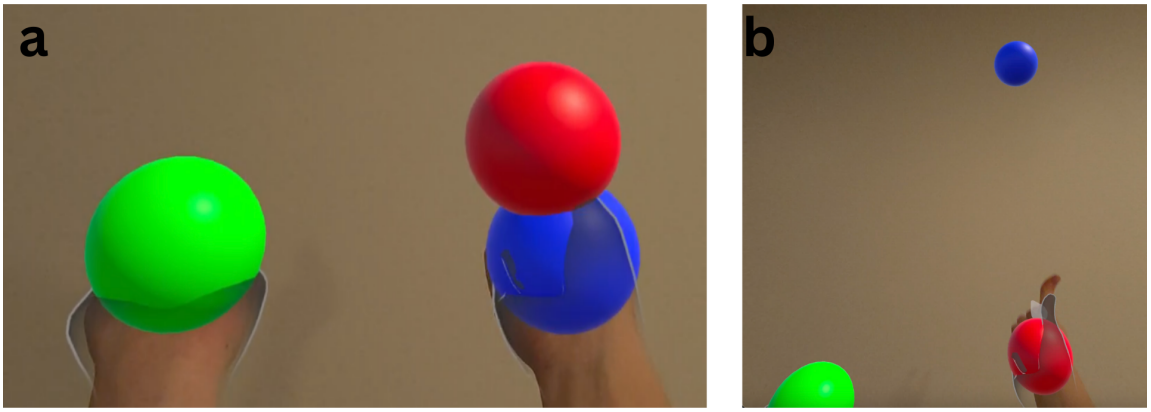


Fig. 2. a) Holding two balls in one hand with a pinching grip for the first throw b) Throwing the first ball from a pinching grip

The project employed an iterative design process to create an application for their user study, utilizing Meta Quest 3 and Unity as the primary development environment. After initial testing among team members, they determined that AR was more suitable than VR for juggling instruction, as VR's total immersion proved more distracting for users.

A key consideration in the AR system was the ball-catching mechanism. While the pinch gesture is common in AR/VR applications, it was deemed unsuitable for developing proper juggling muscle memory. The team recorded multiple hand poses using Meta Quest's hand pose recorder, accounting for various hand sizes within the group. During user studies, participants with hand recognition issues were instructed to use a closed fist gesture, which internal testing confirmed closely mimicked real-life ball-catching mechanics.

The system encountered limitations when handling three-ball juggling scenarios, specifically in recognizing two balls held simultaneously in one hand at the start of juggling. To address this, the team implemented distance grabs for grounded virtual balls using pinch gestures. Users were instructed to first grab virtual balls normally to position them in their palms, then use the pinch gesture to grab the third ball, which would be thrown first. Fig. 2a illustrates how to hold two balls in one hand, with one ball being pinched while the other sits in the palm. Fig. 2b shows how the first ball is then thrown from the pinching position. This ball is the only one thrown from a pinching position, as users are instructed to continue their cycles using regular catching gestures. Subsequently, only prerecorded gestures could be used for mid-air catches. The team validated their system through a single-person formative study prior to the main user study.

3.1 Formative Study

For the formative study, a graduate student was invited to test the AR application for two hours. Throughout the session, the participant provided feedback through structured questioning. Two primary issues emerged from this evaluation. First, the Meta Quest 3's limited field of view required frequent head movement to confirm ball catches, with the participant suggesting that haptic feedback could provide confirmation without visual verification. Second, the participant noted that the hand pose tracking and ball-catching interaction felt artificial, particularly for the left hand.

3.2 Subsequent Design Improvements

To address these concerns, audio feedback was implemented as a substitute for haptic feedback, given the controller-free design of the application. The audio consists of a tennis ball catching sound that plays through the headset upon successful catches. The hand tracking and ball-catching interaction issues were addressed by maximizing the hand tracking frequency on the Meta Quest 3 and expanding the range of hand poses through both additional pre-recorded poses and manually modeled catching positions.

4 User Study

To evaluate the effectiveness, usability, and engagement of AR-based training for learning juggling, we conducted a comparative study between two groups: one trained with traditional physical juggling and the other using an AR environment. The study aimed to investigate how AR facilitates skill acquisition and the transition of muscle memory to physical tasks compared to conventional methods.

4.1 Participants

The study involved eight participants, aged 23 to 29, who were divided evenly into two groups: an AR training group and a physical juggling group. The AR group consisted of two females and two males who used JugglAR, an augmented reality application designed to simulate juggling with adjustable features such as ball speed and gravity. The physical group included three males and one female who practiced juggling with traditional physical balls. All participants had no prior juggling experience and were given the same amount of time across sessions to learn the skill. Progress, skill acquisition, and transitions were monitored to evaluate the effectiveness of each training method.

4.2 Performance Metrics

The primary performance metric for this study is the time and number of sessions that participants need to successfully complete one full juggling cycle (equivalent to two cycles in this study's terminology) with three balls on two separate occasions. This requirement ensures that participants demonstrate consistent juggling proficiency and that both hands are equally engaged, as each of the three balls must transition through both hands during the cycle. This approach aligns with prior research, which highlights the importance of assessing coordination and control over the full juggling cycle when evaluating motor skill acquisition [3]. Using these metrics provides a clear and standardized way to measure progress across both groups.

Participants in both the AR and physical training groups will complete a total of four sessions over two weeks. Each session is designed to monitor progress and assess skill acquisition. The first session is 30 minutes long and includes a 10-minute juggling tutorial video to introduce the basics of juggling, followed by 20 minutes of practice. The subsequent three sessions are 20 minutes each, dedicated entirely to practice.

For the AR-aided group, participants will alternate between practicing in the AR environment and with physical balls after each AR session to ensure alignment with the goal of learning to juggle physical balls. The session number and exact time at which participants achieve the performance metric will be recorded for comparison. This structured approach allows us to evaluate not only the time and consistency of skill acquisition but also the effectiveness of AR in facilitating muscle memory and skill transfer to the physical world.

Table 1. Results of AR Group Participants

Participant	Session Completed	Additional Notes
Participant 1	Session 1 (17 min 32 sec)	Successfully transitioned to physical juggling.
Participant 2	Session 1 (29 min 47 sec)	Successfully transitioned to physical juggling.
Participant 3	Session 2 (11 min 54 sec)	Successfully transitioned to physical juggling.
Participant 4	Session 3 (first 5 minutes)	Successfully transitioned to physical juggling.

Table 2. Results of Physical Group Participants

Participant	Session Completed	Additional Notes
Participant 1	Session 1 (19 min 25 sec)	Successfully juggled with physical balls.
Participant 2	Session 3 (9 min 46 sec)	Successfully juggled with physical balls.
Participant 3	Session 4 (not completed)	Managed only 1 cycle, a decrease from 1.5 cycles in earlier sessions.
Participant 4	Session 3 (not completed)	Could not manage three balls, felt overwhelmed and unmotivated to continue.

4.3 Results and Discussion

The study compared the learning outcomes of participants trained using AR with those practicing juggling solely with physical balls. A total of eight participants were divided into two groups, with four individuals in each group.

As shown in Table 1, all four participants in the AR-based training group successfully transitioned to juggling with physical balls. Participants in this group reported that the AR environment helped them understand the mechanics of juggling and build confidence. However, they also noted challenges related to the limited field of view in the AR system, particularly when looking upward or attempting to catch a ball with a hand that was out of view. While the auditory feedback provided by the application helped mitigate this issue to some extent, the restricted view still made it difficult to fully immerse in the task. Additionally, the absence of tactile feedback, such as the sensation of catching a ball, further impacted the experience. Despite these limitations, all participants in the AR group demonstrated consistent progress and successfully transitioned to physical juggling within three sessions or fewer.

In contrast, Table 2 highlights the more varied outcomes in the physical juggling group. While two participants succeeded within the allotted sessions, one participant was unable to achieve more than one cycle in the final session, showing a regression from earlier sessions. Another participant was unable to juggle with three balls at all, reporting feelings of being overwhelmed and unmotivated to continue. Participants in this group expressed frustration and a lack of enthusiasm, which appeared to hinder their progress and persistence.

These findings suggest that AR-based training facilitated the acquisition of transferable skills for physical juggling, with greater consistency and higher levels of motivation compared to the physical juggling group. The slower progress and lower motivation observed in the physical group underscore the potential advantages of AR-based training in fostering quicker skill acquisition and sustained engagement. However, challenges such as the limited field of view and the lack of tactile feedback in the AR environment highlight areas for future improvement, such as integrating haptic feedback and optimizing the visual experience for a more seamless transition between virtual and physical training environments.

5 Future Work

There are several promising directions for future work to enhance the application and improve its effectiveness in training muscle memory. One potential avenue is the development of gloves with haptic feedback to simulate the sensation of catching a ball [7]. This addition could provide users with real-time tactile feedback, reinforcing learning and increasing the realism of the AR experience.

Another area of improvement involves offering users greater control over the training parameters, such as adjusting the speed, gravity, and number of balls involved in the exercise. By incorporating more flexibility, users can tailor the training experience to match their skill levels and learning preferences. Additionally, the application could be gamified, transforming the experience into an interactive game. This could help determine whether users learn better or faster when the process is framed as a game, potentially increasing motivation and engagement. Prior research suggests that incorporating gamification into VR experiences can significantly improve motor skills related to hand-eye coordination [4].

Technological advancements can also address limitations identified by users, such as the restricted field of view. Upgrading to devices with better field-of-view capabilities would provide a more seamless and immersive experience, reducing barriers that users encountered during testing.

Lastly, a follow-up study is planned to further evaluate the effectiveness of the application. This study will explore whether using AR exclusively during training sessions, followed by transitioning to physical balls at the end, produces similar or superior results compared to allowing users to switch between AR and physical objects throughout the process. This comparative analysis will help refine the application and provide deeper insights into the interplay between AR training and real-world skill acquisition.

6 Conclusion

This study showed that AR-based training effectively aided in the acquisition of juggling skills, with participants transitioning to physical juggling more confidently and consistently than those trained solely through physical practice. The ability to control ball speed and gravity in JugglAR helped participants better understand ball movement and refine their techniques. However, challenges such as the lack of tactile feedback and a limited field of view made the experience less comparable to physical juggling. Despite these drawbacks, the AR group demonstrated a more consistent progress, suggesting that AR can be a valuable tool for teaching complex motor skills, provided its limitations—such as the need for haptic feedback and improved field-of-view—are addressed.

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